

Hybrid Logic Computing of Binary and Stochastic

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Abstract—Binary logic is applied internally to almost all digital signal processing and computer systems, because binary logic is direct implemented in CMOS circuits. Stochastic logic is achieved through its particular representation of data, which uses the probability of the logic level being ON to represent data. Stochastic logic computing is a type of logic computation based on stochastic bit stream instead of the binary numbers. This letter proposes a hybrid computing system of binary logic and stochastic logic, called hybrid logic. The study discusses how to generate hybrid logic circuits, and demonstrates the properties of hybrid logic circuits.

Index Terms—Field-programmable gate array (FPGA), hybrid logic computing, low latency, stochastic computing (SC).

I. INTRODUCTION

THE PROPERTIES of stochastic logic have been the subject of intensive studies for several years. Stochastic logic is a mathematical problem based on probability and mean, and stochastic logic possessing special properties can be easily generated by simple logical circuits to replace complex circuit based on binary logic. On the other hand, in binary logic (Boolean algebra), a variable at any point in the system can assume only one of two values, denoted as 0 or 1.

Conventional digital computing requires much more power consumption when the size of computation is increasing. This causes a tradeoff between the size of the computation and the speed of its solution. Thus, a new data-processing system, stochastic computing (SC), has been considered, which makes the maximum use of every logic gate for computation purposes, and its computations distributed in space and time.

The speed advantage of binary logic in performing digital computations is significant. SC is an important method for pulse computation and was advocated by Gaines in 1969 [1]. In the SC paradigm, computation is performed on stochastic bit stream (SB) instead of binary number (BN). The SB is represented by the probability of firing “1” based on the stream length of the accumulation time [1], [2]. The required computing power of the SC is completely distributed throughout the low-cost logic gate. The binary logic circuit is an efficiently coded representation of data, where the binary logic requires $\log_2 N$ ordered binary levels. Contrarily, SC is the least efficient in the representation of data, which requires N^2 unordered binary levels [1]. However, conventional SC saves considerable hardware resources and power at the expense of efficiency.

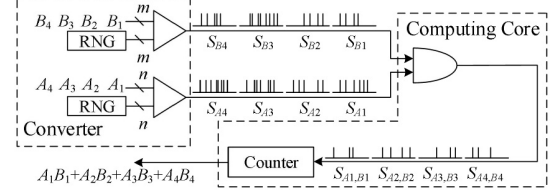


Fig. 1. Classic SC based on SB.

In the SC paradigm, digital computation is performed on SB instead of BN, which has a significant advantage in the occupation of hardware in processing complex functions. SC has been applied to a wide variety of applications [2], [3], such as image processing [4], communication encoding [5], neural networks [6], and deep learning [7]. Sim *et al.* proposed new multiplication methods to reduce stream length and optimize the problem of high latency. Najafi and Lilja [8] proposed deterministic methods to achieve an accurate calculation. Parhi [9] proposed hybrid unipolar and bipolar formats to achieve better energy efficiency. Liu and Han [10] used a Sobol sequence to optimize the problem of poor precision. Canals *et al.* [11] proposed a new encoding method, which uses fractional form to represent a value with two groups of SB and extends the range of pulse representation to the whole real number axis.

High latency is essentially caused by low information carrying ability of 1-bit stochastic stream in conventional SC. When the bit stream length of stochastic logic is equal to N , it can only represent N different data. Clearly, the efficiency of carrying information for bitstream is weaker than that of BN. Some researchers have made some works to improve the efficiency of SC encoding, trying to use multibit stream to reduce latency [12], [13], or using thermometer encoder [14].

High latency and low accuracy are still the major design challenges for the hardware circuits of SC, although SC has the advantages of hardware occupation and noise tolerance. To solve the intrinsic fundamental problem and major design challenges, we proposed a new computing method. The main contributions of this study are as follows.

- 1) Hybrid logic computing of binary logic and stochastic logic is proposed for the hybrid logic SC (HSC).
- 2) The properties and relationship of hybrid, stochastic, and binary logics are discussed.
- 3) The hybrid logic circuit is designed for a multiplier, adder, divider, and the converters of BN and SB.

II. STOCHASTIC COMPUTING

A. Existing SC Methods

The classic multiplication of SC [1] based on a stochastic bitstream is shown in Fig. 1. The stochastic bit stream generator comprises a random number generator (RNG) and comparator. Through the bit stream generator, A and B are converted into stochastic bit streams S_A and S_B , respectively. The probabilities of the two streams being “1” are

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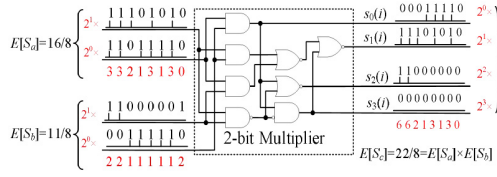


Fig. 2. Multiplication of HSC.



Fig. 3. Addition of HSC.

$P(S_A) = A/2^n$ and $P(S_B) = B/2^m$, respectively. A new stochastic bit stream $S_{A,B}$ is obtained after passing through the “AND” gate. The probability of the output stream being “1” is $P(S_{A,B}) = P(S_A)P(S_B) = AB/2^{n+m}$, which realizes the multiplication of two input streams S_A and S_B . The counter accumulates four stochastic streams $S_{A1,B1}-S_{A4,B4}$ one by one, to obtain the full result of $A_1B_1 + A_2B_2 + A_3B_3 + A_4B_4$.

The SC results may be inaccurate for the following reasons. First, two input streams should be independent; otherwise, $P(S_{A,B}) = P(S_A)P(S_B)$ is false. Second, the stochastic stream is essentially a result of the N_F -fold Bernoulli experiment, and the number of “1” contained in the stream satisfies the binomial distribution [11]. The length of an SB increases exponentially with the desired resolution. For example, to obtain the probability of a stream with 8-bit resolution, one must observe at least 2^{16} bits of stream length using classic SC [15].

B. Hybrid Logic of Stochastic and Binary

SC usually uses the probability of “1” in a stream to represent a value. $x(1), x(2), \dots, x(N)$ represent the stream. Every $x(i)$ in the bit stream is generated by a discrete random variable X with a probability distribution of $P(X = 1) = p$ and $P(X = 0) = 1 - p$, where p is the probability value. According to the definition of expectation in a previous study, the expectation of X can be obtained as follows [17]:

$$E[X] = 1 \cdot p + 0 \cdot (1 - p) = p. \quad (1)$$

According to (1), it can also be regarded as that SC uses the discrete random variable X as a signal source to generate a stochastic stream, and $E[X]$ represents the value of the stream. Because $X = 1$ or 0 in a conventional SC, the stochastic pulse are 1-bit streams.

If SC is understood as described above, the discrete random variable should have a larger range instead of being limited to a 1-bit stream. Therefore, the SC can be extended to use multibit streams instead of 1-bit streams. Examples of the computations on the 2-bit stream are shown in Figs. 2 and 3. Because this new stochastic stream has the characteristics of both conventional stochastic logic and binary logic, we refer to the new method as hybrid logic computing of stochastic and binary, and namely, the multibit stochastic stream as a hybrid stream.

1) *Multiplication and Addition*: If the hybrid streams $s_a(i)$ and $s_b(i)$ are multiplied number by number, a new hybrid stream $s_c(i)$ can be obtained

$$s_c(i) = s_a(i)s_b(i). \quad (2)$$

According to the probability theory if $s_a(i)$ and $s_b(i)$ are independent, the expectation of $s_c(i)$ is the product of $E[S_a]$ and $E[S_b]$, as shown in (3). Therefore, (2) realizes multiplication of the two hybrid streams.

$$E[S_c] = E[S_a S_b] = E[S_a]E[S_b] \quad (3)$$

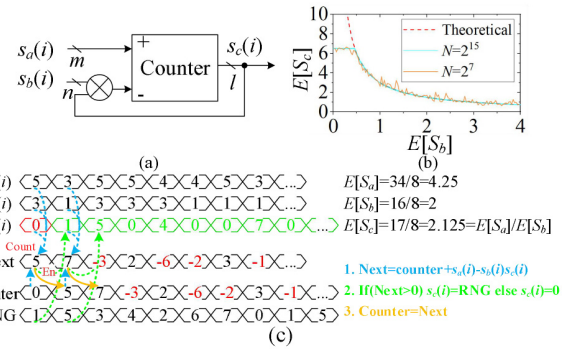


Fig. 4. Division of HSC. (a) Circuit for division computing. (b) Accuracy evaluation of the division circuit, $E[S_c] = 3/E[S_b]$. (c) Example for division circuit.

By adding two hybrid streams $s_a(i)$ and $s_b(i)$ number by number, a new hybrid stream $s_d(i)$ can be obtained

$$s_d(i) = s_a(i) + s_b(i). \quad (4)$$

According to the probability theorem, the expectation of $s_d(i)$ is equal to $E[S_a] + E[S_b]$, as shown in (5). Therefore, (4) realizes the addition of two hybrid streams

$$E[S_d] = E[S_a + S_b] = E[S_a] + E[S_b]. \quad (5)$$

Using Fig. 2 to explain the details, BNs 16 and 11 should first be scaled into $[0, 2^2 - 1]$ to satisfy the representation range of the 2-bit hybrid stream. Therefore, 2-bit streams are encoded with $E[S_a] = 16/8$ and $E[S_b] = 11/8$ to represent 16 and 11, respectively. Using this hybrid logic, n -bit ($n \geq 2$) binary multiplication is realized using a 2-bit multiplier, as shown in Fig. 2. The n -bit ($n \geq 2$) addition is realized using a 2-bit adder, as shown in Fig. 3.

The hybrid logic stream comprises not only 2-bit. When an m -bit hybrid stream is used to represent n -bit binary data ($m \leq n$), n -bit binary multiplication can be operated by an m -bit multiplier.

2) *Division*: Fig. 4(a) shows the division circuit with HSC. Here, $s_a(i)$ and $s_b(i)$ are input hybrid streams, $i = 0, 1, \dots, N - 1$, and the result of division $s_c(i)$ is a hybrid stream. The counter is used to compute $\sum_{i=1}^n [s_a(i) - s_c(i)s_b(i)]$. When the expectation of the counter is equal to 0, $E[S_c S_b] = E[S_a]$ is established. Under this condition, if S_c and S_b are independent, $E[S_c]E[S_b] = E[S_a]$ is obtained. Thus, if the counter is close to 0, $s_c(i)$ approaches the division result. Setting the initial value of counter and $s_c(0)$ to zero and using an RNG to control the value of $s_c(i)$, when counter > 0 , $s_c(i) = \text{RNG}$. When counter < 0 , $s_c(i) = 0$. The accuracy test of the division circuit is shown in Fig. 4(b), $E[S_c] = 3/E[S_b]$, when $E[S_a] = 3$ and $n = m = l = 3$. The data flow of division circuit is shown as Fig. 4(c).

3) *BN-SB Converter*: Two types of BN-SB circuits, Type-A and Type-B, were proposed as shown in Fig. 5. BN-SB(Type-A) uses one BN x (n -bit) as the input to encode the multibit stream. For example, the encoding process for m -bit stochastic stream is shown in Fig. 5(a), where x_i represents the i th bit of x . The low $(n - m + 1)$ -bit of x and RNG is compared to obtain the bit stream $s_0(i)$. The high $(m - 1)$ -bit of x is compared with 1-bit RNGs, respectively, to obtain $s_1(i), s_2(i) \dots s_{m-1}(i)$. Here, 1-bit RNGs can be obtained from an arbitrary bit of the $(n - m + 1)$ -bit RNG. In Fig. 5(a), m is equal to 3 as an example, and 3-bit stream output versus input x has the following relationship:

$$E[S_0] = p_0 = (2^{n-3}x_{n-3} + \dots + 2^1x_1 + x_0)/2^{n-2} \quad (6)$$

$$E[S_i] = p_i = x_{n-3+i}/2, i = 1, 2. \quad (7)$$

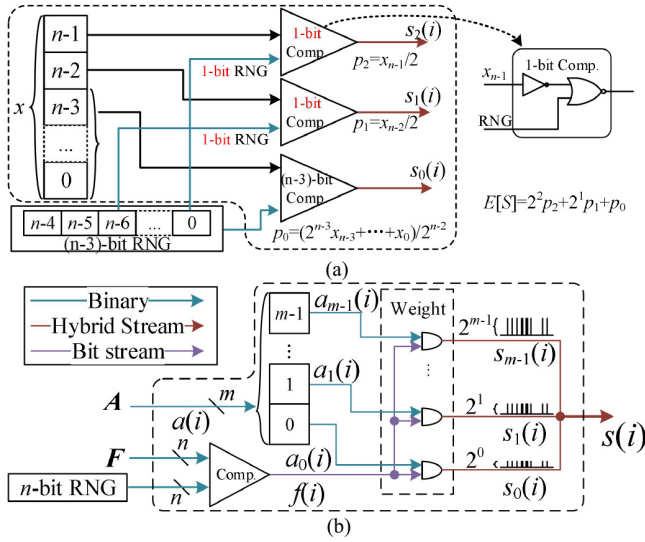


Fig. 5. BN-SB converter for hybrid logic computing. (a) Type-A uses one binary data to encode 3-bit stream. (b) Type-B uses two binary data to encode hybrid stream.

Thus, the expectation of the 3-bit stream output by BN-SB (Type-A) is proportional to x , which realizes multibit stream encoding for BN x

$$E[S] = 2^2 p_2 + 2^1 p_1 + 2^0 p_0 = \left(2^{n-1} x_{n-1} + 2^{n-2} x_{n-2} \dots + x_0 \right) / 2^{n-2} = x / 2^{n-2}. \quad (8)$$

Type-B, illustrated in Fig. 5(b), is another method for encoding multibit streams, which uses two BNs to encode a multibit stream. The output multibit stream represents the product of two input BNs. The BN $x(n+m)$ -bit to be encoded is the product of y and p , where $x = y \cdot p = y \cdot z / 2^n$. z is an n -bit, y is an m -bit, and $p \in [0, 1]$. The bit stream $f(i)$ representing p is generated by the comparison between z and a random number, which is the same as the BN-SB circuit for a conventional SC

$$E[F] = p = z / 2^n. \quad (9)$$

Thereafter, if $a(i) = y$ and $i = 1, 2, \dots, N$, then $E[A] = y$.

Finally, $s(i)$ is obtained by multiplying $a(i)$ and $f(i)$, as shown on the right-hand side of Fig. 5. Here, $a_0(i), a_1(i), \dots, a_{m-1}(i)$ indicate m bits of $a(i)$. m bits of $s(i)$ have the same representation as $a(i)$

$$s(i) = \sum_{j=0}^{m-1} 2^j s_j(i) = \sum_{j=0}^{m-1} 2^j [a_j(i) \cdot f(i)]. \quad (10)$$

The value expressed by $s(i)$ is determined by the expectation of $s(i)$. Because A is a constant, the expectation is obtained as shown in (11). Therefore, $s(i)$ is a multibit stream representing x , and the circuit achieves the purpose of converting the BN x into a multibit stream

$$E[S] = \left(\sum_{j=0}^{m-1} 2^j y_j \right) E[F] = y \cdot E[F] = y \cdot p = x. \quad (11)$$

Hybrid logic computing has a lower requirement for stream length N than conventional bit stream to achieve the same resolution. When converting an n -bit BN x to an $(m+1)$ -bit stream, it is sufficient to have an $(n-m)$ -bit resolution ($N = 2^{n-m} - 1$) for every bit stream of the $(m+1)$ -bit stream to represent x , where an n -bit resolution ($N = 2^n - 1$) is required for classic SC.

When $m = 0$, the hybrid logic stream degrades into a classic bit stream. It is an extreme situation that has the lowest

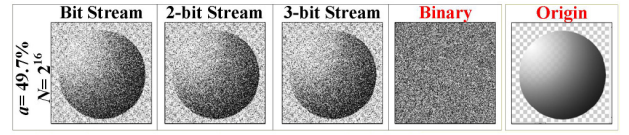


Fig. 6. Experiment for noise tolerance of HSC.

hardware occupation but the worst latency. When $m = n - 1$, the hybrid logic stream is converted into a BN. This is another extreme situation that has the best latency and largest hardware occupation. Through this hybrid logic, latency and hardware occupation can be traded off instead of choosing between two extreme situations.

III. DISCUSSION FOR HYBRID LOGIC COMPUTING

A. Noise Tolerance of Hybrid Logic

In a noisy environment, it is assumed that every bit of stochastic stream has the same bit flip rate $a \in [0, 0.5]$. In addition, a bit stream $x(i)$ is generated by the discrete random variable X , and the expectation of X is $p \in [0, 1]$. X_{noise} is used to represent random variables affected by noise. The four situations for the bit flip of $x(i)$ are as follows.

- 1) Send 1 get 1, $p_1 = p(1 - a)$.
- 2) Send 1 get 0, $p_2 = pa$.
- 3) Send 0 get 1, $p_3 = (1 - p)a$.
- 4) Send 0 get 0, $p_4 = (1 - p)(1 - a)$.

Thus, the expectation of X_{noise} is obtained as follows:

$$E[X_{\text{noise}}] = p_1 + 0 \cdot p_2 + p_3 + 0 \cdot p_4 = (1 - 2a)E[X] + a. \quad (12)$$

The characteristics of SC illustrate that there is a determined linear relationship between the expectation of X_{noise} and X , which is related to the bit flip rate a . This characteristic gives the SC a high noise tolerance. It should be noted that when $a = 0.5$, $E[X_{\text{noise}}] = 0.5$, X_{noise} has no relationship to X , and the information is completely lost.

The m -bit stream $s(i)$ can be seen as a parallel of bit streams $s_0(i), \dots, s_{m-1}(i)$, and (13) is also applicable to $s_0(i), \dots, s_{m-1}(i)$. Assuming that every bit of stochastic stream has the same bit flip rate a and $E[S_i] = p_i, i = 0, 1, \dots, m - 1$. The expectation of S is

$$E[S] = 2^{m-1} E[s_{m-1}] + 2^{m-2} E[s_{m-2}] + \dots + 2^0 E[s_0]. \quad (13)$$

According to (12) and (13), the expectation of the m -bit stream $s(i)$ affected by noise is given by (14). The bit flip error also leads to a linear change in the expectation of the hybrid stream, which makes it easy to recover the original data. Thus, the hybrid stream has the same noise tolerance as that of the bit stream. The noise tolerance of HSC is shown based on bitstream, multibit stream, and binary data in Fig. 6.

$$\begin{aligned} E[S_{\text{noise}}] &= 2^{m-1} E[s_{m-1, \text{noise}}] + \dots + 2^0 E[s_{0, \text{noise}}] \\ &= (1 - 2a) \left(2^{m-1} p_{m-1} + \dots + 2^0 p_0 \right) + a \left(2^{m-1} + \dots + 2^0 \right) \\ &= (1 - 2a) E[S] + a(2^m - 1). \end{aligned} \quad (14)$$

B. Accuracy Experiment for Hybrid Logic

Owing to the existence of progressive precision [16], accuracy can easily be changed by choosing different stream lengths, which is one of the advantages of SC. Because the hardware implementation of a true RNG is more complex, a pseudorandom sequence is usually used in an SC. When using a pseudorandom sequence, obtaining progressive precision through statistics is a better strategy, and it is used in the following text.

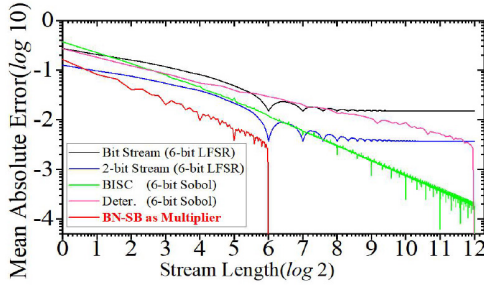


Fig. 7. MAE of classic SC [1], BISC [7], Deterministic [15], 2-bit stream, and BN-SB for HSC.

TABLE I
LATENCY(N), AREA, POWER, AND EFFICIENCY OF THREE METHODS

Type of logic		Bit Stream			Hybrid Stream		Binary
Method		[1]	[7]	[15]	Fig.2	BN-SB	
RNG (n -bit)		LFSR	Sobol	Sobol	LFSR	Sobol	
Bit Width (n -bit)	4	MAE	0.03	0.03	0.03	0.03	0
		N	16	17	81	9	1
	6	MAE	0.017	0.017	0.016	0.017	0
		N	64	46	169	39	1
	8	MAE	0.008	0.008	0.008	0.008	0
		N	256	122	576	169	1
Area ($n=8$)	LUT	18	29	20	20	17	64
	FF	32	40	32	32	24	48
*($\times 10^{-4}$), $n=8$		2.17	2.8	0.87	2.95	24.5	156.3
Power (W) ($n=8$, 600 MHz)		0.009	0.01	0.009	0.008	0.007	0.018
**(OPS/W)($\times 10^9$)		0.26	0.49	0.12	0.44	3.57	33.3

*Area Efficiency=(LUT $\times N$) $^{-1}$, N is the stream length, which represents latency.

**Energy Efficiency=600 MHz/(N ·Power), $n=8$ -bit, OPS means the number of multiplication operations per second.

Using binary computation as the benchmark, the deviation in multiplication between the SC and the NB was calculated. The mean absolute error (MAE) is used as the accuracy metric (a lower MAE means a higher accuracy). $SC(A, B, N)$ is the result of SC for $A \times B$ with stream length N , and $P(A, B)$ is the theoretical value for $A \times B$, where A is n -bit and B is m -bit. When $MAE(N) = 0$, the result of SC is completely consistent with the BN calculations, indicating that a zero-error calculation is achieved [15]

$$MAE(N) = \frac{\sum_{A=0}^{2^n-1} \sum_{B=0}^{2^m-1} |SC(A, B, N) - P(A, B)|}{2^{n+m}}. \quad (15)$$

Fig. 7 shows the MAE of the four methods with different RNGs when performing 6-bit multiplication. Among the bit stream methods, classic [1], the binary-interfaced SC (BISC) [7] and clock division of deterministic method [15], BISC has the best progressive precision. BN-SB has a speed increase of 2^6 times that of deterministic to achieve zero-error calculation. The MAE of the 2-bit stream method is better than that of bit stream methods when $N < 2^6$.

Table I shows the latency of hybrid logic computing and conventional SC to reach the same precision, using MAE to measure the error. As mentioned in [8], the generated bit stream repeats and so the accuracy of classic method [1] does not improve after this stream length $N > 2^n$. Therefore, the stream length N in the test is also no more than 2^n . Based on the best accuracy achieved by the classic SC operating on BNs with different bit widths, the stream length of other methods that achieve the same accuracy is tested.

The circuits for different methods with 8-bit data input were implemented using Kintex-7 FPGA (field-programmable gate array), and all of the data are gained from design tool, as shown in Table I. (LUT $\times N$) $^{-1}$ was counted to measure the

area efficiency of each method. BN-SB as a multiplier and BISC both have the limitation that they can only be used in the first multiplication of polynomial operations, or that the stochastic stream output needs to be converted to BNs before the next multiplication. The area efficiency of BN-SB as a multiplier is the best among the methods in the test, which is 11.3 times that of the classic SC method. The 2-bit hybrid stream and BISC have a close area efficiency, which is approximately a 30% increase in area efficiency compared with the classic SC.

IV. CONCLUSION

This study presents a new data representation method called HSC. The proposed method uses the expectation of multi-bit streams to achieve SC. Compared with conventional SB, hybrid logic computing realizes low-latency and low-area occupation instead of the long latency of the classic method. We demonstrated the rationality of computing rules for multi-bit streams and designed the basic circuit for an HSC. In the experiments of precision, using BN-SB as a multiplier reduced the latency by a factor of $1/2^m$ to reach zero error calculation. With 8-bit input data, the area efficiency of BN-SB as a multiplier was 11.3 times that of the classic SC method.

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